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RESEARCH ARTICLE

A Scoping Review of Research on Designers and GenAI: Who Uses What, and What For?

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ABSTRACT:

Generative artificial intelligence (GenAI) is increasingly integrated into design practice, yet it remains unclear which designers use these technologies, which tools they adopt, and for what purposes. This scoping review examines empirical research on professional designers' engagement with GenAI, published between January 2020 and December 2025. Specifically, it analyzes 25 peer-reviewed journal articles and conference papers: 20 identified through Scopus and Web of Science, and five through a targeted journal search. The findings reveal considerable diversity and ambiguity in how design professions are defined, reflecting the interdisciplinary and evolving nature of contemporary design practice. Designers working across physical, digital, and service-oriented contexts are represented, often with overlapping disciplinary labels. Publicly available GenAI tools - especially ChatGPT, Midjourney, DALL-E, and Stable Diffusion - predominate and are frequently used across multiple stages of the design process, while only a few studies report custom-built tools. Efficiency emerges as a unifying motivation, with GenAI accelerating workflows, automating repetitive tasks, and enabling rapid iteration. Creative exploration and ideation are also prominent, with GenAI framed as a co-creative partner. This review clarifies the research landscape and highlights gaps for future study.

Keywords: Artificial Intelligence; generative AI; GenAI tools; design; scoping review

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Introduction

Generative artificial intelligence (GenAI) has rapidly emerged as a groundbreaking force in creative and design disciplines, introducing capabilities that enhance design efficiency, foster innovative solutions, and expand the boundaries of artistic exploration [1]. The rapid expansion of this research area warrants mapping its current developments [2]. To this day, recent studies inform us on many aspects. Through the automation of complex design tasks, GenAI acts as a collaborative partner, empowering designers to explore and iterate on creative ideas at an unprecedented pace and scale [3]. In design practice, GenAI systems are no longer mere tools but active collaborators that help designers address complex requirements and unlock new creative potential [4]. In this respect, a recent review by Mirzaei et al. [5] indicates that AI is increasingly integrated into design thinking processes, broadening ideation and accelerating prototyping and testing, while human judgment remains central to selection and ethical decision-making. However, the implications of these

developments are not uniform across design practice. The field of design encompasses a wide range of disciplines, practices, and professional contexts that extend from industrial and product design to digital and service design [6]. This diversity makes design a particularly dynamic and interdisciplinary domain, yet it also complicates understanding how GenAI tools are being adopted across its many subfields. While GenAI's role in creative and artistic practices has been widely discussed, there remains limited clarity about which types of designers are actively using GenAI tools and what specific tools they employ (e.g., ChatGPT, DALL-E, Midjourney, or other domain-specific systems).

Through this scoping review, I aim to provide an overview of the design landscape regarding the use of GenAI, identify the designers and GenAI tools most frequently discussed in the literature, and highlight which design disciplines remain underexplored. The review maps and describes how GenAI is represented in current design research and practice, focusing on who the designers are who engage with GenAI tools and on the types of tools explored across different areas of design practice. It also examines the purposes for which designers implement such technologies. Overall, the review offers an exploratory, rather than evaluative, synthesis that clarifies which design fields and GenAI tools have been studied most extensively and where research gaps persist. The following three research questions guide this review:

- **RQ1**: Which types of designers are empirically studied in relation to the use of GenAI tools?
- **RQ2**: Which GenAI tools are examined or introduced in empirical studies involving designers?
- **RQ3**: For what purposes are GenAI tools used or explored by designers within these studies?

The research questions are intentionally mapping-oriented, consistent with the purpose of a scoping review: "to map rapidly the key concepts underpinning a research area and the main sources and types of evidence available," often "undertaken as stand-alone projects in their own right, especially where an area is complex or has not been reviewed comprehensively before" [2,7]. Two structural characteristics of the field further highlight the need for a scoping review.

First, design research is marked by persistent terminological ambiguity, with identical labels used to denote substantively different practices. For example, product design has traditionally referred to the development of physical, manufactured artifacts [8], yet the same term is increasingly used to describe the design of digital products such as mobile applications and software services [9]. Second, the ecosystem of GenAI tools is evolving at an exceptional pace, particularly following the public release of large language models such as ChatGPT, with tools frequently emerging, being discontinued, or rebranded; Google Bard's later renaming as Gemini is one example of this instability [10]. Both of these dynamics make it difficult to establish a stable overview of who is using which GenAI tools and for what purposes, thereby necessitating a scoping review that maps current empirical research rather than synthesizing outcomes or effectiveness.

Understanding who implements GenAI tools and which tools are most prevalent is an important first step in grasping how artificial intelligence is shaping contemporary design practice. Mapping this landscape provides the foundation for analyzing how different areas of design respond to and evolve with these emerging technologies. The following section outlines the current state of research on the impact of AI within design workflows, offering contextual insights into how generative systems are transforming creative processes across diverse design domains.

1. Background

1.1. Who Is a Designer?

Design is a field that is not easy to define. As Ambrose and Harris explain, "[a] designer can fulfill various roles in different environments and company structures" [11]. Even though the International Standard Classification of Occupations categorizes designers into "Product and Garment Designers," "Graphic and Multimedia Designers," and "Interior Designers and Decorators" as an umbrella, categorizing design disciplines within those categories is not straightforward [12]. Product design, for instance, has traditionally referred to the creation of physical artifacts, yet in recent years, the term has also been applied to digital contexts. Russell and Buck [8] define product design as a professional practice concerned with the creation of physical, tangible, and manufactured artifacts produced at scale. In contrast, Chung et al. [9] use the term in relation to digital products (specifically mobile applications), highlighting how product design principles are increasingly employed in software and digital service development. Lastly, design itself can also be seen as a process of change through which designers communicate and materialize that change [13]. This lack of terminological stability complicates attempts to synthesize empirical findings across studies and obscures comparisons between design disciplines, reinforcing the need for reviews that explicitly map how design roles are defined and operationalized in the literature.

1.2. Defining Artificial Intelligence and GenAI

The term artificial intelligence (AI) was first introduced in the 1950s as "the simple theory of human intelligence being exhibited by machines" [14]. Recent technological advancements have made this theory a reality through developments in machine learning, deep learning, and generative models, which enable machines to analyze data, recognize patterns, and generate increasingly sophisticated outputs [15]. Nevertheless, the concept of AI lacks a universally agreed-upon definition. As Schmager et al. observe, the term has been applied in diverse ways, from describing a machine capable of 'learning and thinking' to more recent definitions that present AI as a machine-based system that produces outputs such as predictions, recommendations, or decisions [16]. What these diverse perspectives share, nonetheless, is an emphasis on the relationship between AI systems and human interaction, highlighting the socio-technical nature of AI (meaning that AI systems cannot be separated from the social contexts, stakeholders, and human interactions in which they are developed, deployed, and used).

While AI serves as an umbrella term, this article focuses on GenAI, which primarily refers to the subfield of AI focused on generating new content in various formats, such as text, images, videos, and code, as the term "generative" implies [15]. García-Peñalvo and Vázquez-Ingelmo define GenAI "as the production of previously unseen synthetic content, in any form and to support any task, through generative modeling" [17].

1.3. The Impact of GenAI on Design Practice

In recent years, GenAI integration has brought transformative tools to many design disciplines, boosting creativity and workflow for designers and artists [1]. GenAI functions both as a creative tool and a collaborative partner, helping designers tackle complex projects more effectively, much like the printing press or photography [18]. This development, however, adds layers of complexity, as designers must manage stakeholder expectations and refine GenAI outputs through iterative processes [19]. According to Bendoly et al. [20], companies skilled in GenAI, particularly in additive manufacturing, demonstrated notable adaptability during the COVID-19 pandemic by using human-AI collaboration to innovate and solve unexpected problems. This flexibility fosters double-loop learning, advancing design practices. GenAI's impact extends across fields such as game design, sound production, jewelry and textile design, and educational craft applications, emphasizing its potential to transform design with greater precision, efficiency, and co-creation [21-25]. In addition, the rapid evolution of GenAI tools, including frequent updates, discontinuities, and rebranding [10] poses challenges for maintaining a coherent, up-to-date understanding of which tools are examined across empirical studies.

2. Methods

2.1. Scoping Literature Review

This scoping review follows the methodological framework developed by Arksey and O'Malley [2], which provides a rigorous and transparent approach to mapping existing literature across a broad topic. A scoping review was selected rather than a systematic literature review because the field of designers' use of GenAI remains evolving and conceptually fragmented, making it more appropriate to map the breadth of existing evidence and identify research gaps than to synthesize or appraise effectiveness [26, 27]. Unlike systematic literature reviews, which aim for exhaustive synthesis and theoretical consolidation, scoping reviews are designed to explore and map the breadth of an emerging field [2, 28].

The framework consists of five key stages: identifying the research question, identifying relevant studies, study selection, charting the data, and collating, summarizing, and reporting the results. Arksey and O'Malley [2] also proposed a sixth, optional consultation stage in which researchers engage relevant stakeholders (e.g., practitioners, policymakers, or affected groups) to validate and enrich the review's findings. Later, Levac et al. [29] refined this framework, recommending that the consultation phase should be treated as an integral part of the scoping review process. However, as this review is conducted by a

single author with limited access to external stakeholders, it will focus on the five core stages of the original framework. The steps are outlined in more detail below.

2.1.1. Identifying the Research Questions

The first stage of a scoping review involves defining a clear and focused research question, as this step guides all subsequent stages [2]. To formulate the research questions, a preliminary exploration of the literature was conducted to refine the scope of inquiry and ensure feasibility. This initial exploration helped assess the breadth of available literature, confirm that similar reviews had not already been conducted, and verify that sufficient studies exist to warrant a scoping review. Such preliminary testing of keywords is a useful step in clarifying research boundaries and ensuring that the review remains both manageable and relevant [26].

2.1.2. Identifying Relevant Studies

To identify relevant studies, this scoping review employed a two-phase search strategy. In the first phase, searches were conducted in Scopus and Web of Science (WoS) using the following query: ("design practition*" OR "design profession*") AND ("GenAI" OR "Generative AI" OR "Artificial Intelligence"). The search strategy deliberately relied on the umbrella terms "design practitioner" and "design professional" to avoid imposing a priori disciplinary boundaries on what constitutes a designer. Given the well-documented ambiguity and overlap of design roles, the use of broader descriptors allowed the review to capture how studies themselves define professional designers, rather than pre-classifying them through predefined labels [13]. Following the approach of Schmager et al. [16], this review relies on Scopus and WoS as the primary databases for literature retrieval. As demonstrated by Mongeon and Paul-Hus [30], Scopus provides broader journal coverage across academic disciplines, making it a robust foundation for comprehensive searches. To enhance completeness and identify studies not indexed in Scopus, a complementary search in WoS yielded additional relevant publications. This dual-database strategy supports balanced and systematic coverage across fields while maintaining methodological transparency. The search was restricted to peer-reviewed journal articles and conference papers to ensure the inclusion of academically validated and high-quality research.

In the second phase, a targeted journal search was conducted, encompassing all Level 2 (top-tier) design journals classified in the Norwegian Register for Scientific Journals. These included *Computer Aided Geometric Design; Computer-Aided Design; Crystal Growth and Design; Design and Culture; Design Issues; Design Studies; Formal Methods in System Design; International Journal of Design; Journal of Design History; Journal of Engineering Design; Research in Engineering Design; and Road Materials and Pavement Design*. In addition, the targeted search included *Design and Artificial Intelligence*, a journal launched in 2025 with a specific focus on topics directly aligned with the scope of this review. **Figure 1** illustrates the literature selection process in a flow diagram inspired by the PRISMA guidelines [31].

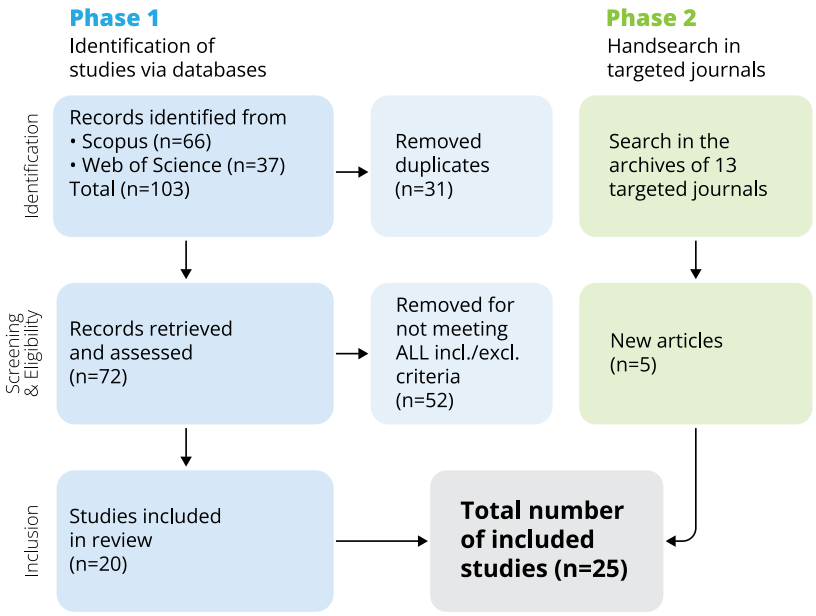


Figure 1. PRISMA-inspired flow diagram illustrating the literature selection process. Figure drawn by author.

Table 1. Final corpus of 25 identified papers. Table drawn by author.

Alharthi [34]	Kamath et al. [22]	Queiroz [51]
Anggrianto [35]	Lee and Park [43]	Süner-Pla-Cerdà et al. [52]
Ansari [36]	Li et al. [44]	Wang et al. [53]
Baquero and Almeida [37]	Li et al. [45]	Wärnestål [54]
Chen and Xu [38]	Lin et al. [46]	Weisz et al. [55]
Cheng [39]	Ma et al. [47]	Yang [56]
el Kordy [40]	Ma et al. [48]	Zhang et al. [57]
Engawi et al. [41]	Mannan and Islam [49]	
Hasanah and Lindblom [42]	Obieke et al. [50]	

2.1.3. Study Selection

The search strategy was created based on specific inclusion criteria. Articles were eligible for inclusion in this scoping review if the terms "design practitioner" or "design professional" appeared alongside "GenAI," "Generative AI," or "Artificial Intelligence" in the title, abstract, and/or keywords. Only studies published between January 2020 and December 2025 (December being the last month in which the search for papers took place) were included to ensure the contemporary relevance of GenAI technologies in design practice. The publications were limited to peer-reviewed journal articles and conference papers written in English. Additionally, only empirical studies presenting original data (whether qualitative, quantitative, or mixed methods) were considered; opinion pieces,

theoretical commentaries, reviews, and editorials were excluded. The included studies had to involve professional designers and not university students studying to become designers. A preliminary systematic exploration of platforms like Google Scholar indicated that there are many more studies on university students than on practicing professionals [32]. Similar observations have been made for other disciplines, for example, prospective and practicing teachers [33]. The current review focuses on professionals who practice design and implement GenAI tools in their daily work. These criteria ensured that the selected studies provided evidence-based insights into how design professionals engage with GenAI tools across various domains of practice. These selection criteria resulted in a final corpus of 25 identified papers, all of which are presented in **Table 1**.

2.1.4. Charting the Data

This phase involved systematically extracting and organizing key information from each included study to enable comparison across the corpus. In line with Arksey and O'Malley's [2] "descriptive–analytical" approach, I created a structured charting form to capture both study-specific details, including author(s), year, design discipline(s) involved, GenAI tools examined, and the stated purposes of implementation. Each article was read carefully to record terminology related to design professions, the specific GenAI tools mentioned or used, and the rationale for their application within design practice. This process allowed the data to be grouped thematically around the review's three research questions (who the designers are, what tools they use, and why). The charting process was iterative and reflexive, meaning that as familiarity with the literature grew, categories were refined and new subthemes were added to ensure comprehensive coverage and internal consistency.

2.1.5. Collating, Summarizing, and Reporting the Results

In this final stage, the extracted data were collated and organized to provide both a descriptive overview and a thematic summary of the reviewed literature. Following Arksey and O'Malley's [2] guidance, the aim was not to assess the quality of evidence but to map the scope, trends, and gaps within the field. The data were synthesized through a descriptive–analytical approach, grouping studies according to the three research questions above. Patterns were identified across design disciplines and tool types, and recurring themes were developed to capture the drivers of adoption, including efficiency, creativity, and reflection. This process resulted in an evidence-based summary that outlines the current state of research on designers' engagement with GenAI and highlights areas where empirical inquiry remains limited.

2.2. Methodological Limitations

While every effort has been made to capture the full breadth of research on designers' use of GenAI tools, several limitations should be acknowledged. The search strategy focused on studies explicitly referring to "design practitioners" or "design professionals," which may have excluded research that identified participants using discipline-specific titles, such as "fashion designers." Consequently, some design subfields may be underrepresented in the dataset. The review also included only English-language, peer-reviewed publications, potentially limiting the inclusion of gray literature or studies published in other languages.

As this was a single-author review, the selection and charting processes were conducted independently, which may have introduced interpretive bias despite efforts to apply inclusion criteria systematically and transparently. Furthermore, the deliberate focus on identifying design disciplines represented in GenAI-related research and the specific tools used within those disciplines over the past five years may have introduced selection bias. Studies addressing AI from broader or alternative design perspectives, such as those centered on ethical or human-centered design without a direct focus on GenAI tools, were excluded. For example, Schmager and Sousa examine the design of trustworthy AI through the development of a trustworthy AI design toolkit and related ethical considerations, but do not empirically investigate designers' engagement with GenAI tools, and therefore fall outside the scope of this review [58].

3. Findings and Discussion

Rather than seeking to theorize or evaluate GenAI adoption in design, this review adopts a mapping-oriented perspective, examining how the field is currently represented across empirical studies with regard to who uses these technologies, which tools are employed, and for what purposes. Consistent with the aims of scoping reviews, the objective is to clarify and organize an emerging and conceptually fragmented body of work rather than to produce an exhaustive or theoretically consolidated account [2, 26, 27]. In the present section, findings and discussion are integrated to present the empirical patterns identified in the literature while critically engaging with their meaning and implications. The paper therefore offers a descriptive synthesis of empirical research published between 2020 and 2025, while simultaneously highlighting the conceptual ambiguities, practical tensions, and broader concerns that emerge across the reviewed studies. Across the included corpus, three overarching observations stand out: the diversity and ambiguity of design roles, the distribution of GenAI tools across design domains, and a consistent emphasis on creativity and efficiency as central motivations for adoption.

3.1. Which Types of Designers Are Empirically Studied in Relation to the Use of GenAI Tools?

As explained earlier, defining who qualifies as a designer is inherently complex because design encompasses a diverse range of practices and contexts. The findings reveal a wide variety of design professions using or being introduced to GenAI tools, as shown in **Table 2**. This variety reflects both the richness of the design field and the challenge of establishing clear disciplinary boundaries. For example, two of the identified disciplines, visual communication design and graphic design, demonstrate this ambiguity. The term visual communication design is often used more broadly to describe practices that convey meaning and information through visual media, such as advertising, publishing, and other forms of mediated communication [59]. However, in some contexts, the term is used interchangeably with graphic design; as Barnard [59] notes, the two disciplines share considerable conceptual overlap. While Anggrianto [35] explicitly uses the term visual communication designers, it is recorded here exactly as it appears in the study, as the author does not clarify

whether this refers specifically to graphic designers, making it a useful example of how disciplinary terms can overlap and remain context-dependent. Additionally, some studies that focus on a specific design discipline employ generic terminology when referring to their participants. For example, two papers frame their respective work around industrial design/designers [52] and interior design/designers [36], yet describe their participants using the broader labels professional designers and design professionals. The absence of participant role specification also highlights the need for caution when interpreting disciplinary boundaries in empirical findings. Overall, the range of disciplines identified, from UX and product design to sound, architecture, and visual communication, demonstrates the multifaceted nature of contemporary design and the cross-disciplinary adoption of GenAI across its various subfields. Additional design disciplines were identified in the broader literature, such as fashion design [32]; however, these studies were not included in the final corpus because they were conducted with university students rather than practicing design professionals.

Table 2. Design professions. Table drawn by author.

Design professions	Source paper(s)
Architects	[43], [48], [49]
Brand specialists	[41]
Digital designers / artists	[51]
Digital media designers	[48]
Engineering designers	[50]
Game designers	[34]
Graphic designers	[37], [41], [46], [48]
Industrial designers	[48], [52], [53], [57]
Interactive designers	[48], [53], [55]
Interior designers	[36], [48]
Product designers	[38], [45], [53]
Service designers	[48], [54]
Software designers	[55]
Sound designers	[22]
UI designers	[42], [48]
User researchers	[42], [55]
UX designers	[42], [44], [48], [54]
Visual communication designers	[35]
Visual designers	[48], [55]

One of the most prominent findings of this review is the extent to which design practice is described through diverse and sometimes inconsistent terminology. As noted earlier, this ambiguity partly reflects designers' adaptability, which often involves moving across different environments and types of problem-solving activities [11]. This openness can be understood as a strength of the field, as it allows designers to work across a wide range of projects and contexts without being confined to rigid professional or educational boundaries. At the same time, from a research perspective, such flexibility also creates challenges. When roles are described inconsistently, it becomes more difficult to compare studies systematically and to draw clear conclusions about how particular groups of designers engage with GenAI tools.

This tension between the openness of design practice and the need for analytical clarity points to the value of a more pragmatic comparative framework. Based on the mapping and synthesis conducted in this review, an outcome-oriented categorization of design practice is proposed as a useful lens for comparing heterogeneous studies. Specifically, three broad outcome-oriented categories can be distinguished: designers producing physical products (e.g., industrial, jewelry, furniture), designers producing digital products (e.g., digital user interfaces and mobile applications), and designers engaged in service design. This categorization was derived inductively from the corpus of this review; therefore, disciplines with distinct epistemological traditions, such as speculative design [60] and participatory design [61], may not fit neatly into outcome-oriented frames. Additionally, as noted in Section 2.2, studies that identified participants by discipline-specific titles, such as fashion designers, may have been excluded by the search strategy, leaving certain design subfields likely underrepresented within these categories. At the same time, this categorization does not replace existing disciplinary labels but instead offers a practical way to compare GenAI adoption across diverse design practices. These categories should not be understood as mutually exclusive, as many designers work across multiple outcome domains and position their practice at the intersections of physical, digital, and service-oriented design. Future researchers are therefore encouraged to apply this categorization reflexively, remaining open to adjusting it to fit the specific needs of each study, so as to embrace the diversity and richness of the design field.

3.2. Which GenAI Tools Are Examined or Introduced in Empirical Studies Involving Designers?

The reviewed studies indicate that designers employ or are introduced to a diverse range of GenAI tools, selecting them according to the specific needs and stages of their design projects. For example, Li et al. [44] demonstrate how different tools are integrated across phases of a UX project: Midjourney, Stability Photoshop Plugin, and ChatGPT are used during the early stages of design; Notion AI supports the drafting of design proposals; Prompt2Design assists in generating UI components; and MiroAI helps organize ideation notes. Such findings illustrate that designers strategically combine multiple GenAI systems rather than relying on a single platform. While the majority of tools identified across the literature are publicly available (e.g., ChatGPT, Midjourney, and Stable Diffusion), a small number of custom-built tools were also observed. For instance, Kamath et al. [22]

developed bespoke audio Creative Support Tools powered by StyleGAN-based generative models, which were empirically studied with professional sound designers to support the interactive generation and manipulation of audio content within their creative practice. This mix of commercial and custom systems reflects a dynamic landscape in which designers adapt GenAI technologies to different creative contexts and project requirements.

Numerous models have been proposed to describe the design process. See, for example, Wynn and Clarkson [62], who provide a detailed analysis of different models. Nonetheless, the terms *analysis*, *synthesis*, and *evaluation*, as described by Martinec et al. [63], are used here as a practical framework for mapping where the identified GenAI tools have been applied across the design process. Simply put, *analysis* refers to the understanding, decomposition, and formulation of problem-related entities, such as requirements, constraints, and needs. *Synthesis* is concerned with generating and exploring solution alternatives, such as ideas, concepts, and working principles. *Evaluation* assesses solution entities, judging their suitability and triggering further refinement. It should be noted, however, that this mapping is partial and approximate, as not all reviewed studies describe GenAI usage in terms of explicit process stages. **Table 3** provides an overview of existing and custom-built GenAI tools identified in these reviewed studies, listing the source papers in which each tool is mentioned and, where possible, the stage of the design process to which its use relates. It should be noted that six of the identified papers [35, 36, 40, 41, 54, 56] are not included in **Table 3**, as they do not include any specific GenAI tools in their empirical part. Rather, they explore, among other things, designers' dispositions towards the use of such tools more generally.

These findings further suggest that designers tend to adopt GenAI opportunistically, integrating publicly available tools such as ChatGPT, Midjourney, DALL·E, and Stable Diffusion into existing workflows rather than reorganizing practice around a single system. Designers frequently combine multiple tools across different stages of the design process, from early ideation to prototyping and refinement [44]. This pattern may be closely linked to the diversity of design as a field, where practitioners work across varied domains, processes, and creative priorities. Because designers often approach problems from fundamentally different perspectives, no single GenAI tool appears sufficient to meet the full range of professional needs. The use of multiple, general-purpose systems can therefore be understood as a pragmatic response to the heterogeneous character of design practice. However, it may also indicate that existing GenAI tools remain only partially aligned with the specific requirements of different design disciplines, encouraging adaptation around available systems rather than deeper integration into professional workflows. Nevertheless, to clarify the relationship between tool use and design profession, **Table 4** provides a cross-disciplinary overview of the GenAI tools reported or examined in the reviewed studies. Two types of design professionals—brand specialists [41] and visual communication designers [35]—are not presented in **Table 4**, even though they are included in **Table 2** The reason is that the studies in which these professionals are included do not specify any GenAI tools.

Table 3. GenAI tools observed. Table drawn by author.

		Stages where tools are used		
		Analysis	**Synthesis**	**Evaluation**
Existing GenAI tools	Adobe Firefly and/or Adobe Photoshop (Generative Fill)		[38], [51], [55]	[38], [51], [55]
	AIVA		[55]	[55]
	Artbreeder		[34]	
	ChatGPT	[34], [42], [43], [44], [45], [46], [47], [51], [53], [55]	[34], [37], [42], [44], [45], [47], [51], [55]	[44], [47], [50], [53], [55]
	ControlNet	[52]	[45], [52]	[45], [46], [52]
	DALL-E	[45], [52]	[34], [38], [44], [45], [47], [50], [51], [52], [55]	[38], [45], [47], [52], [55]
	DeepSeek			[50]
	DreamStudio	[52]	[52], [55]	[52]
	Figma plugins		[42]	
	Gemini (previously Google Bard)	[34]	[34], [55]	[55]
	GitHub Copilot		[55]	[55]
	Grok			[50]
	IBM watsonx.ai Prompt Lab			[55]
	Meshy		[53]	[53]
	Microsoft Copilot	[47], [50]	[34], [47], [51]	[47], [50], [51]
	Midjourney	[38], [39], [45], [52]	[34], [38], [39], [42], [44], [45], [46], [47], [48], [51], [52], [53], [55]	[38], [39], [44], [45], [52], [55]
	Miro AI	[44]		
	Notion AI		[44]	
	Prompt2Design		[44]	
	Scenario		[34]	
	Soundraw		[34]	
	Stability Photoshop Plugin (0.7.0)		[44]	
	Stable Diffusion	[38], [45]	[34], [38], [43], [44], [45], [46], [48], [53]	[38], [45], [46]
	Vizcom	[52]	[52]	[52]
Custom-built GenAI tools	Custom style-perception prompt system using Stable Diffusion	[38]	[38]	[38]
	Custom AI-based Audio Creative Support Tools (StyleGAN-based)	[22]	[22]	
	Custom IEDS system tool (based on Stable Diffusion XL 1.0)		[57]	
	Pro-Explora	[50]	[50]	
	Vector Quantized Generative Adversarial Network (VQGAN) together with a pre-trained CLIP model		[49]	[49]

Table 4. Mapping of GenAI tools observed across design professions. Table drawn by author.

	Architects	Digital designers / artists	Digital media designers	Engineering designers	Game designers	Graphic designers	Industrial designers	Interactive designers	Interior designers	Product designers	Service designers	Software designer	Sound designers	UI designers	User researchers	UX designers	Visual designers
Adobe Firefly and/or Adobe Photoshop (Generative Fill)		×						×		×		×			×		×
AIVA								×				×			×		×
Artbreeder					×												
ChatGPT	×	×		×	×	×	×	×		×		×		×	×	×	×
ControlNet						×	×			×							
DALL-E		×		×	×		×	×		×		×			×	×	×
DeepSeek				×													
DreamStudio							×	×				×			×		×
Figma plugins														×	×	×	
Gemini (previously Google Bard)					×			×				×			×		×
GitHub Copilot								×				×			×		×
Grok				×													
IBM watsonx.ai Prompt Lab								×				×			×		×
Meshy							×	×		×							
Microsoft Copilot		×		×	×												
Midjourney	×	×	×		×	×	×	×	×	×	×	×		×	×	×	×
Miro AI																×	
Notion AI																×	
Prompt2Design																×	
Scenario					×												
Soundraw					×												
Stability Photoshop Plugin (0.7.0)																×	
Stable Diffusion	×		×		×	×	×	×	×	×	×			×		×	×
Vizcom							×										
Custom GenAI tools	×			×			×			×			×				

3.3. For What Purposes Are GenAI Tools Used or Explored by Designers Within These Studies?

In this section, purpose refers to the underlying rationale or motive for implementing GenAI tools in design practice, rather than to the stage of the design process in which a tool is used. While Section 3.2 maps where GenAI tools are applied across design process stages such as *analysis*, *synthesis*, and *evaluation*, this section examines why designers use or explore these tools. This paper concludes that designers use or explore GenAI tools for three broad and recurring purposes identified across the corpus: expanding ideation and creative exploration, enhancing efficiency, and, less frequently, supporting learning, adoption, and capability-building. In keeping with the exploratory nature of a scoping review, these categories are intended to provide an overview of the main patterns in the literature rather than an exhaustive account of every specific purpose reported in individual studies.

In ideation and creative exploration, designers use GenAI to stimulate creativity and overcome design blocks by generating diverse, unexpected visual outputs, particularly in early-stage conceptual design [34]. Studies describe GenAI as a co-creative partner that broadens the design space and supports experimentation [45, 52]. In architectural and industrial contexts, GenAI imagery serves as a conceptual visualization tool to convey early ideas and atmospheres [43] and to help designers explore new stylistic directions and aesthetic combinations [52]. This exploratory role appears particularly strongly in studies of product, architectural, and engineering design, where GenAI is used to generate alternative concepts, speculative forms, and unexpected combinations during divergent phases of the design process [49, 50, 53, 57]. Similarly, in game and sound design, GenAI is framed less as a tool for final execution than as a creative partner that helps designers rapidly explore unconventional ideas, atmospheres, and novel outputs [22, 34].

In terms of efficiency, productivity, and iteration speed, designers use GenAI to automate repetitive or time-consuming tasks, such as generating visuals, text, or layouts [42, 44]. The tools enable faster prototyping and accelerate design iteration by shortening the feedback loop between ideation and execution [45]. In UX contexts, GenAI supports the rapid creation of interface variations and content, allowing designers to test multiple options quickly [44]. Efficiency-oriented uses are especially visible in UX, graphic, and product design contexts, where GenAI is used to automate wireframing, generate layout or rendering variations, and reduce the time required for routine production tasks [37, 38, 44]. At the same time, even studies that support efficiency rarely frame GenAI as a productivity tool alone. Rather, time saved through automation is often presented as creating more room for designers to focus on higher-level creative or strategic work [40, 41]. Collectively, these findings suggest that efficiency-oriented and creativity-oriented uses of GenAI are not mutually exclusive but often co-exist within design practice, with designers moving fluidly between productivity gains and exploratory experimentation. Gaining insight into whether designers primarily seek efficiency gains or creative augmentation helps determine where to position future studies. For example, it could be considered whether to focus on productivity and iteration processes, on ideation and creative exploration, or to adopt a

cross-disciplinary perspective that explicitly integrates both. Beyond creativity and efficiency, some studies suggest that GenAI can also support capability-building among designers, particularly in relation to learning, emerging forms of AI literacy, and shared understanding within multidisciplinary teams [47, 54].

Although less prominent than ideation and efficiency, this third cluster of studies points to a growing concern with how designers learn to work with GenAI, what shapes their willingness to adopt such tools, and how professional roles may shift as these technologies become embedded in practice. The identified studies emphasize this specifically in their conclusions and recommendations for further research [see 35, 48, 56]. Conceptual and exploratory work further frames GenAI use as raising important questions about designers' computational understanding and literacy, highlighting capability development as a key concern for future practice [51, 64].

Design practice has traditionally been organized around five core purposes: problem definition, ideation, user research, prototyping, and testing [5]. These purposes are not a rigid sequence but an iterative, interdependent set of activities that together structure the creative and problem-solving process across design disciplines. Two of the three GenAI purposes identified in the current review (i.e., ideation and creative exploration, and efficiency) map directly onto this tradition. Ideation has long been central to design practice, and efficiency concerns have shaped prototyping and iteration since the earliest process models. What GenAI changes is not the purpose itself but its character: the speed and scale at which ideation can be pursued, and the shift in creative initiation from the designer alone to a human-machine collaboration in which GenAI proposes and the designer selects, refines, and evaluates [5]. The third purpose identified here (i.e., capability-building and AI literacy) has no clear parallel in the pre-GenAI empirical design literature as a primary motivation for professionals. Learning has always been part of design practice, but the specific concern with understanding and governing AI tools as they are embedded into workflows represents a qualitatively different kind of professional development, one driven by the need to maintain agency over an increasingly active collaborator rather than simply to acquire a new skill.

Nevertheless, not all traditional design purposes are being transformed by GenAI, and some appear to remain largely irreplaceable by current tools. Empathic user research—understood as sustained, relational engagement with users' lived experiences, tacit needs, and contextual realities—depends on forms of interpretation, trust-building, and situated judgment that are not readily replicated by data synthesis or automated analysis [5]. Similarly, stakeholder negotiation, ethical judgment, and final aesthetic decision-making continue to require human accountability and contextual sensitivity [65, 66, 67]. As Mirzaei et al. [5] conclude, while AI can broaden the evidence base for empathy and accelerate ideation and prototyping, human judgment remains central to selection, values-driven choices, and ethical decision-making. These enduring human responsibilities are not residual functions left over after automation; they are the constitutive core of design practice, and the GenAI purposes identified in this review are best understood as augmenting the surrounding work rather than displacing its center.

The emphasis on efficiency, as identified here, spans physical, digital, and service-oriented design contexts, suggesting that productivity gains are a foundational driver of GenAI adoption in design practice. At the same time, this strong focus on efficiency brings into view a set of broader and often less explicitly addressed questions about design work and working conditions. If designers are expected to produce more in less time, this may gradually reshape professional expectations, workplace norms, and perceptions of adequacy within design practice. In a fast-paced AI-enabled context, taking longer to develop ideas or outputs may increasingly be seen as inefficiency rather than as a necessary part of thoughtful design work. These concerns also connect to wider debates about automation and employment, including questions about role compression, job displacement, and the restructuring of design labor [see, e.g., 68]. If one designer, supported by GenAI, can carry out work previously distributed across several people, this raises important questions about how organizations may respond and how the profession itself may change.

At the same time, creativity-oriented uses of GenAI should not be understood as separate from efficiency-oriented ones. As noted above, designers use GenAI to stimulate creativity, overcome design blocks, and explore unexpected visual, textual, or conceptual directions [34]. In this sense, GenAI is framed as both an automation tool and a co-creative partner that expands the range of possible design outcomes [for more, see 69]. Yet this creative support may itself be enabled by efficiency gains: by accelerating routine production tasks, GenAI may free up more time for experimentation, brainstorming, and the development of original ideas. The rapid expansion of GenAI in design practice is therefore transforming what designers do, how they work, what is expected of them, and the conditions under which creative work takes place.

Overall, these developments suggest that the increasing use of GenAI is reshaping design environments in ways that call for critical attention to their ethical and professional implications. Questions of responsibility, fairness, and sustainability become increasingly important as such tools are embedded in everyday practice. While some studies also point to other motivations, such as learning, reflection, and capability-building, these appear less frequently and are often framed as emerging benefits rather than as primary reasons for adoption.

Conclusion

The findings of this scoping review highlight both the opportunities and challenges of examining GenAI use in contemporary design practice. The diversity and inconsistency of design roles and terminologies across studies complicate cross-study comparison; however, an outcome-oriented categorization of designers offers a pragmatic approach for synthesizing research across heterogeneous design contexts. However, this framework may be of limited utility for design disciplines whose practice is not primarily defined by outcome type, such as speculative design or participatory design. Researchers working in or across such fields are encouraged to adapt or extend the categorization to better reflect the particular logics of their discipline. With respect to tools, the review reveals a clear

dominance of publicly available, general-purpose GenAI systems that designers integrate flexibly into existing workflows rather than adopting as single, all-encompassing solutions. While a small number of studies document the development of custom-built GenAI tools, these systems remain the exception and tend to emerge in response to specific domain requirements that are not adequately supported by commercial platforms. Across disciplines, efficiency emerges as a unifying motivation for GenAI adoption, with designers consistently emphasizing gains in productivity, speed, and iteration. These efficiency-oriented uses frequently coexist with creative and exploratory practices, in which GenAI is framed as a co-creative partner rather than a purely automating technology. Taken together, the review provides a descriptive foundation for future research by clarifying who is using GenAI, which tools are most prevalent, and why they are adopted. Further empirical investigation into underexplored design domains, domain-specific GenAI systems, and the longer-term implications of efficiency-driven adoption may help to better understand how GenAI is reshaping design practice, both in practical terms and in relation to designers' professional roles and identities.

Implications for Future Research

Although GenAI appears to offer substantial practical value for designers across multiple levels of practice, including proofreading, ideation, rapid prototyping, and routine production tasks, its implications should not be assessed only in terms of immediate usefulness. Future research would benefit from examining GenAI adoption more systemically, with greater attention to how these tools may reshape design work, professional expectations, and organizational conditions over time. While the current review shows that efficiency and creative exploration are dominant motivations for adoption, broader structural consequences remain insufficiently understood.

Further empirical research is therefore needed to investigate unmet needs across underrepresented areas of design practice and to explore how domain-specific GenAI systems might address requirements that general-purpose tools do not adequately support. At the same time, little is known about how designers' motivations for using GenAI evolve as these technologies become more deeply embedded in everyday practice. Longitudinal and ethnographic studies could offer important insight into whether efficiency-driven adoption results in lasting changes to workflows, professional roles, and creative agency, or whether the perceived benefits stabilize, shift, or diminish over time.

Future studies should also examine the wider workplace implications of GenAI integration. As these tools make it possible to produce more in less time, expectations around speed, productivity, and professional adequacy may change accordingly. This raises important questions about what may happen to design work in environments where slower, more reflective processes risk being devalued. Related concerns about job displacement, role compression, and labor redistribution within design teams also warrant closer investigation. Understanding these developments is essential for assessing the practical impact of GenAI and for evaluating its ethical and professional consequences.

All in all, these issues suggest that future research should move beyond documenting where and why GenAI is used, and increasingly address how its adoption is transforming the conditions under which design is practiced. Such work would help develop a more critical and systematic understanding of GenAI in design, including how these technologies can be implemented responsibly and in ways that support both innovation and sustainable professional practice.

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